Adarsh Dodeja

Buffalo, NY (open to relocation) | adarshdodeja15@gmail.com | github.com/Adarsh8966

Profile

Robotics and Mechanical Engineer with dual B.S. degrees in Aerospace and Mechanical Engineering. Builds and tests robots end to end, from CAD and custom chassis design to embedded firmware, ROS 2 controls and SLAM. Analytical, creative problem-solving approach backed by test data: 90%+ obstacle-avoidance success over 50 runs and 99% detection accuracy over 100 controlled experiments. Strong attention to detail in technical documentation, wiring and prototype validation. Authorized to work in the U.S. on OPT; no sponsorship required.

Technical Skills

Engineering Tools: SolidWorks, Fusion 360, Autodesk Inventor, KiCad, CAD/FEA, DFM and tolerancing, 3D printing, soldering, oscilloscope/DMM troubleshooting, MATLAB, XFLR5, RDS-Win
Robotics & Embedded: ROS 2, Gazebo, RViz, SLAM Toolbox, `robot_localization` EKF, nonlinear MPC (CasADi/IPOPT), inverse kinematics, encoder-feedback control, Raspberry Pi, nRF52840, Arduino/ESP32, SPI, I^2C, CAN, UART, BLE
Software: C++, Python, Linux, Docker, OpenCV, NumPy/pandas, Git
Strengths: analytical, creative, problem-solving, attention to detail, technical documentation, test data analysis, experiments
Certifications (in progress): FAA Part 107 Remote Pilot; Lean Six Sigma

Education

University at Buffalo (SUNY) Buffalo, NY
B.S. Aerospace Engineering and B.S. Mechanical Engineering (Dual Degree) • Dean's List
Coursework: Mechatronics, Propulsion Engineering, Aircraft Design, Fluid Mechanics, Numerical Methods, Manufacturing Processes, Mechanics of Solids, Engineering Materials, Structural Analysis

Experience

Robotics & Automation Internship Jun 2024 – Aug 2024, Jun 2025 – Aug 2025 • Mumbai, India
Industrial Robotic Arm Integration • Biotech manufacturing facility
- Integrated an existing 6-DOF robotic arm for automated pick-and-place of manufacturing components: wired SPI hardware, implemented CAN-bus communication and control software, and configured a 100 Hz closed-loop servo loop.
- Taught and programmed pick-and-place motions using inverse kinematics, reaching 0.5 mm positional repeatability when returning repeatedly to the same taught position.

Operations & Accounting Assistant May 2023 – Jan 2026 • Mumbai, India
DS Traders
- Supported inventory and replenishment across 10 retail locations (60–100 SKUs, about $75,000/month in transactions) and built Python data-verification workflows for purchasing reconciliation and reporting, with careful attention to detail.

Projects

Autonomous Mecanum Rover (ROS 2, Controls, SLAM) Aug 2026 – Present
- Designed a custom laser-cut chassis in Fusion 360 and built a ROS 2 Jazzy stack on a Raspberry Pi 5: mecanum controller (50 Hz wheel commands), `robot_localization` EKF, 2D LiDAR SLAM and nonlinear MPC (CasADi/IPOPT) for tracking and obstacle avoidance.
- Cut straight-line heading error 63% with encoder-feedback velocity control versus the initial open-loop controller on the same robot.
- Achieved at least 90% obstacle-avoidance success over 50 runs with under 500 ms response, and 8 cm mean return-to-start error over 20 tape-measured trials.
- Reached 95% target-detection accuracy on 100 evaluated images under varying lighting, with camera inference offloaded to a laptop GPU over Wi-Fi.

Wearable Baseball Pitch Tracker (Senior Design, Team Lead) Jan 2026 – May 2026
- Led the team and wrote the embedded C++ firmware on an nRF52840 with a BMI323 IMU: signal filtering, event-state logic and BLE telemetry; designed the PCB in KiCad, hand-assembled the working electronics on perfboard and contributed to the SolidWorks enclosure.
- Reached 99% pitch-detection and 95% velocity accuracy across 100 controlled experiments; presented the prototype to 100+ attendees at the senior design showcase.

Counter-Drone Fiber-Optic Tether Cutting Mechanism (UAV Hardware) Sep 2025 – May 2026
- Developed a drone-mounted cutting mechanism with camera-based wire localization, path planning and an FPV headset for human-guided cutting; iterated three prototypes in Autodesk Inventor and Fusion 360 with CAD/FEA and tolerance/DFM decisions.
- Diagnosed a servo limited to about 180°, found the restricting gear, modified it and verified roughly 360° range; supported 15+ UAV field tests.

3D-Printed Air Glider (MAE 339 Aircraft Competition)
- Designed and 3D printed an unpowered airframe, taking 1st place in the class with a 120-foot flight, the longest recorded.

Additional

Conceptual Design & Analysis: Military STOL transport (5+ configurations, RDS-Win/XFLR5/MATLAB); ramjet cycle performance modeling (MATLAB/Python); biogenic insulation trade study with decision matrix. **AIAA Student Chapter:** Model rocketry and weather-balloon payload integration.